

From Fig. 12, Since $R_c = 2 \text{ k}\Omega$ and $R_L = 10 \text{ k}\Omega$
The effective load Z_L for Amplifier

$$= R_c \parallel R_L = \frac{2 \text{ K} \times 10 \text{ K}}{2 \text{ K} + 10 \text{ K}} = \frac{20 \text{ K}}{12} = 1.67 \text{ k}\Omega$$

$$A_1 = \frac{-h_{fe}}{1 + h_{oe} \cdot Z_L} = \frac{-50}{1 + 25 \times 10^{-6} \times 1.67 \times 10^3} = -48$$

$$Z_{in} = h_{ie} + A_1 \cdot h_r \cdot Z_L$$

$$= 10^3 - 48 \times 4 \times 10^{-4} \times 1.67 \times 10^3$$

$$= 1000 - 32 = 968 \Omega = 0.968 \text{ k}\Omega$$

$$A_v = \frac{A_1 \cdot Z_L}{Z_{in}} = \frac{-48 \times 1.67 \times 10^3}{0.968 \times 10^3} = -82.8 \approx -83$$

$$Y_o = h_{oe} - \frac{h_{fe} \times h_{re}}{R_S + h_{ie}} = \frac{h_{oe} \cdot R_S + h_{oe} \cdot h_{ie} - h_{fe} \cdot h_{re}}{R_S + h_{ie}}$$

$$\text{if } \Delta h = h_{ie} \cdot h_{oe} - h_{fe} \cdot h_{re}$$

$$Y_o = \frac{h_{oe} \cdot R_S + \Delta h}{R_S + h_{ie}}$$

$$Z_o = \frac{R_S + h_{ie}}{h_{ie} \cdot R_S + \Delta h} = \frac{2 \times 10^3}{3 \times 10^{-2}}$$

$$= 67 \times 10^3 = 67 \text{ k}\Omega$$

$$Z_o'' = \frac{67 \times 10^3 \times 1.67 \times 10^3}{68.67 \times 10^3} = 1.63 \times 10^3$$

$$= 1.63 \text{ k}\Omega \quad (\because Z_o'' = Z_o \parallel Z_L)$$

$$Z'_1 = \frac{Z_1 \cdot R_B}{Z_1 + R_B} = \frac{0.968 \times 10^3 \times R_B}{0.968 \times 10^3 + R_B} \rightarrow (R_B = R_1 \parallel R_2)$$

$$R_B = \frac{100 \times 10^3 \times 10 \times 10^3}{100 \times 10^3 + 10 \times 10^3} = \frac{10^9}{110 \times 10^3}$$

$$= \frac{1000 \times 10^3}{110} = 9.1 \text{ k}\Omega$$

$$Z'_i = \frac{0.968 \times 10^3 \times 9.1 \times 10^3}{9.1 \times 10^3 + 0.968 \times 10^3} = \frac{8808.8}{10.068} = 875 \, \Omega$$

$$= R_i \rightarrow \left(\because Z'_L = \frac{Z_{in} \cdot R_B}{(Z_{in} \cdot R_B)} \right)$$

$$A_{vs} = \frac{V_o}{V_s} = \frac{V_o}{V_i} \times \frac{V_i}{V_s} = \frac{V_o}{V_i} \times \frac{R_i}{R_s + R_i}$$

$$= A_v \times \frac{R_i}{R_s + R_i} = \frac{-83 \times 0.875 \times 10^3}{1 \times 10^3 + 0.875 \times 10^3} \approx -39$$

A_{vs} refers to voltage gain taking the signal source resistance R_s into consideration and A_{is} refers to current gain taking R_s into consideration.

$$A_{is} = A_i \times \frac{R_s}{R_s + R_i} = -48 \times \frac{10^3}{1.875 \, K} = 25.6 \approx -26$$